GLOWTAP

Standard Database Performance Benchmark

Industry latency SLOs & standard test-data volumes for OLTP SaaS (bookings / dashboards / writes).

Industry norms only | 6 Aug 2026 | Postgres OLTP

Document type: Industry reference (latency SLOs & test-data norms)

Audience: Product / engineering / stakeholders

Date: 6 August 2026

Scope: OLTP SaaS databases (bookings, dashboards, writes) — *not* warehouse/BI-only workloads

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1. Purpose

This document defines **industry-typical** performance targets and **dataset sizes** used when benchmarking transactional Postgres-style applications (e.g. salon/booking platforms).

Use it to answer:

1. What latency range is considered **good**?

2. How much data should performance tests usually load?

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2. Latency standards (what “good” means)

Times below are **end-to-end query / API latency** (includes network when remote). Prefer **p95** for product SLOs; **avg/p50** for developer benches.

2.1 By query type

WORKLOAD	EXCELLENT (LOCAL / SAME-REGION)	GOOD	ACCEPTABLE	INVESTIGATE
Point lookup (PK / unique key)	≤ 5–10 ms	≤ 20–40 ms	≤ 100–150 ms	> 200–300 ms
Filtered list (paginated, indexed)	≤ 20 ms	≤ 50 ms	≤ 150–200 ms	> 300–500 ms
Multi-table join (screen-sized)	≤ 50 ms	≤ 100 ms	≤ 250–300 ms	> 500 ms
Single-row INSERT / UPDATE	≤ 10–20 ms	≤ 50–80 ms	≤ 200–300 ms	> 500 ms
Analytics / platform rollup (interactive)	≤ 200 ms	≤ 1 s	≤ 2–3 s	> 3–5 s
Analytics (batch / offline OK)	—	< 30 s	< 5 min	job design

2.2 Product-level API SLOs (common industry practice)

METRIC	TYPICAL TARGET
Interactive API **p50**	< 100 ms
Interactive API **p95**	< 200–300 ms
Interactive API **p99**	< 500 ms – 1 s
User-blocking UI	Keep under ~2–3 s; else async / pagination

2.3 Local vs remote (serverless) interpretation

ENVIRONMENT	WHAT THE NUMBER MOSTLY REFLECTS	HOW TO READ IT
Local Postgres	Disk / CPU / planner / indexes	Treat **≤ 10–40 ms** lookups as strong
Remote serverless (e.g. Neon)	**WAN RTT + pooler +** DB work	Lookups often sit **~80–150 ms** even when the plan is fine

A remote lookup of ~100 ms can be **healthy** if EXPLAIN shows index use and local equivalent is a few ms.

A local lookup of ~100 ms on an indexed unique key usually means **missing index, cold disk, or bad plan**.

2.4 Thumb-rule “good range” (quick reference)

- **Lookups / lists / writes (local):** aim **under ~50 ms**; celebrate **under ~10–20 ms**
- **Lookups (remote same-ish region):** aim **under ~100–150 ms**
- **Dashboard aggregates (interactive):** aim **under ~200–500 ms**; move heavier work async
- **Anything interactive over ~2–3 s:** redesign query, add index, or make it a background job

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3. Standard test-data volumes

Industry performance tests are only trustworthy when hot tables are large enough that **wrong plans hurt**.

3.1 Recommended dataset sizes

STAGE	HOT-TABLE ROWS (APPROX.)	OVERALL DB	USE
CI / smoke	10K – 100K	< 500 MB	Correctness + basic timings
Engineering perf (standard)	**1M – 10M+** on hottest tables	**~1–10 GB**	Index, join, pagination realism
Pre-prod / capacity	10M – 100M+ or production clone	tens of GB+	Peak load, degradation
Analytics / warehouse	Months–years of history	Separate store often	Reporting SLOs, not OLTP

3.2 Content rules (not just row count)

1. **Skew:** some orgs/branches huge, some tiny (like production).

2. **Indexes under load:** unique keys, `(org_id, status)`, date ranges, FKs used in joins.

3. **Writes included:** INSERT / UPDATE / DELETE — not SELECT-only.

4. **Warm-up + N rounds:** discard first run; report avg / p50 / p95 from measured rounds.

5. **Separate environments:** local CPU baseline vs remote RTT-inclusive baseline — never mix grades blindly.

3.3 Rule of thumb

> If a full table scan of the hottest table would still finish in tens of ms, the dataset is **too small** for a serious OLTP perf claim.

Use at least **~1 million rows** on primary transactional tables (appointments, attendance, ledger, etc.) for release-grade perf statements.

4. Methodology checklist (industry-aligned)

STEP	STANDARD PRACTICE
Warm-up	1 discarded run (cache / cold start)
Measured rounds	3–10 timed runs (5 is common)
Reported stats	min, avg, **p50**, **p95**, max
Grades	Band by workload type (lookup vs write vs analytics)
Environment note	CPU/RAM or compute CU, region, DB size, row counts
Plans	Spot-check with `EXPLAIN (ANALYZE, BUFFERS)` on outliers
Pass criteria	No interactive path in **Investigate** band on target env

5. Grading bands used in GlowTap suite (aligned to §2)

These bands encode the industry ranges above for automated reports.

Local PostgreSQL

GRADE	LOOKUPS / JOINS	WRITES	ANALYTICS
EXCELLENT	≤ 10 ms	≤ 20 ms	≤ 200 ms
GOOD	≤ 40 ms	≤ 80 ms	≤ 1 s
ACCEPTABLE	≤ 150 ms	≤ 300 ms	≤ 3 s
SLOW	above	above	above

Neon / remote serverless

GRADE	LOOKUPS / JOINS	WRITES	ANALYTICS
EXCELLENT	≤ 30 ms	≤ 50 ms	≤ 500 ms
GOOD	≤ 100 ms	≤ 200 ms	≤ 2 s
ACCEPTABLE	≤ 400 ms	≤ 800 ms	≤ 5 s
SLOW	above	above	above

6. One-page summary

QUESTION	STANDARD ANSWER
Is my lookup “good”?	Local **< 20–40 ms**; remote **< 100–150 ms** usually yes
Is my interactive dashboard “good”?	Prefer **< 200–500 ms**; treat > 2–3 s as a product problem
How much data to load for real perf?	**1M–10M+ rows** on hot tables; full suite often **multi-GB**
Smoke data enough?	No — fine for CI, not for “we’re production-ready” claims
What to publish?	Environment, dataset size, rounds, **p95**, grade mix, top slow queries

7. References (practice basis)

- Interactive SaaS API SLO practice: p95 often in the **200–300 ms** band
- Postgres OLTP tuning focus: index-backed point lookups and keyset pagination under **millions of rows**
- Serverless Postgres: separate **RTT** from **execution time** via EXPLAIN vs client wall clock
- GlowTap automated suite: `SchemaMaking/superAdmin/scripts/run-perf-benchmark.js`

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